

Soil Fertility Atlas Bangladesh 2020

Curated Extract for Embeddings

Prologue • Background • Methodology • National Soil Fertility Status Tables

This file is a selective extract of the Soil Fertility Atlas Bangladesh 2020 (SRDI, 36pp.), prepared for a RAG knowledge base. It includes the narrative sections (Prologue, Background, Methodology) and all 12 national soil-fertility status tables (area and percentage of arable land by fertility class, for pH, organic matter, phosphorus, potassium, sulphur, zinc, boron, calcium and magnesium). The 12 choropleth soil maps and 12 pie charts in the source are excluded: the maps are raster images with no embedded text layer (their legend/area figures are not machine-readable), and the pie charts visualize the exact same figures already present in the tables below, so including them would only duplicate data. Every fact below was verified against the source PDF's actual text layer, not transcribed from the visual layout. Table numbering: the source PDF mislabels its final two tables as "Table 10" (a typo, reused for Boron, Calcium and Magnesium alike); they have been renumbered 10, 11 and 12 here to match the document's own front-matter List of Tables. Citation: SRDI, 2020. Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3). pp.36.

Prologue

The first national soil fertility maps were prepared by Soil Resource Development Institute (SRDI) in 1998. These maps were prepared on the basis of soil nutrient data generated during Reconnaissance Survey (RSS) and semi-detailed soil survey for the preparation of upazila-wise “Land and Soil Resources Utilization Guides”, popularly known as “Upazila Nirdeshika”. Soil fertility maps include phosphorus, potassium, sulfur, zinc, boron, organic matter and pH status maps. These maps were incorporated in “Fertilizer Recommendation Guide-2005”.

Second soil fertility maps, including phosphorus, potassium, sulfur, zinc, boron, organic matter and pH status maps, were prepared in 2010. These maps were prepared on the basis of soil nutrient database generated for the preparation of Upazila Nirdeshika. These maps were also incorporated in “Fertilizer Recommendation Guide-2012” and “Fertilizer Recommendation Guide-2018”. Phosphorus status map was made only for Loamy to Clayey Soils of Wetland Rice Crops, considering critical limits and fertility class for modified Olsen method (neutral to calcareous soils) values and Bray & Kurtz method (acid soils) values. Potassium and sulfur status maps were prepared for Loamy to Clayey Soils of Wetland Rice Crops. Zinc and Boron status maps were prepared considering the usual soil fertility class described in “Fertilizer Recommendation Guide- 2005, 2012 and 2018”. Soil Reaction (pH) and Soil Organic Matter (SOM) status maps were also prepared, considering classification described in the Fertilizer Recommendation Guides.

There is an urgent necessity to have new soil fertility maps by National Agricultural Research System (NARS) scientists and also by Bangladesh Agricultural Research Council (BARC). These maps were used in Agro-Ecological Zone (AEZ)-wise fertilizer recommendation for major crops and cropping patterns of that particular AEZ, for estimation of fertilizer requirements and distribution of fertilizers, and in location-specific soil fertility management.

Understanding the necessity of updated national soil fertility maps, the “Strengthening of Soil Research and Research Facilities (SRSRF)” project under SRDI took initiative to prepare new national-level soil fertility maps. SRDI collected soil samples during semi-detailed soil survey conducted for preparing Land and Soil Resources Utilization Guides for every upazila of the country, at a sample intensity of at least one sample per 200 hectares of land (intensity increases with differences in physiography, agro-ecological zones, mapping units, land type and soil group). Collected soil samples were analyzed for nitrogen, phosphorus, potassium, sulfur, zinc, boron, calcium, magnesium, iron, copper and manganese, and for soil pH, organic matter and soil salinity. This chemical data was recorded in Upazila Nirdeshika (hard copy) and digitized in a GIS platform. The first round of soil survey and Upazila Nirdeshika publication ran from 1985 to 2002; the second round is ongoing, and so far 245 Upazila Nirdeshika have been published with chemical data digitized in GIS.

Present Soil Fertility Maps 2020 are based on soil chemical data from Upazila Nirdeshika updated up to 2018. The maps cover: phosphorus (Loamy to Clayey Soils of Upland Crops), phosphorus (Loamy to Clayey Soils of Wetland Rice Crops), potassium (Loamy to Clayey Soils of Upland Crops), potassium (Loamy to Clayey Soils of Wetland Rice Crops), sulfur (Loamy to Clayey Soils of Upland Crops), sulfur (Loamy to Clayey Soils of Wetland Rice Crops), zinc, boron, calcium, magnesium, pH and organic matter status. A nitrogen status map was not prepared because nitrogen status is uniformly very low to low across the entire country. These maps may be further updated as new data becomes available.

These maps are intended to be useful for fertilizer management planning, procurement, distribution, and broad fertilizer recommendations. They may also inform the next Fertilizer Recommendation Guide to be prepared by BARC, and help select research topics and research sites.

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), Prologue, pp.04-05]

Background

As a first step of graduation from developing to developed country while ensuring food security, the necessity of a sustainable and environment-friendly agricultural system is imperative. To make Bangladesh food-sufficient, the agriculture sector has been given highest priority. The government is developing agriculture in consideration of the 7th Five-Year Plan, the National Agriculture Policy, and the SDGs. As per Vision 2021, food sufficiency was targeted by 2013; despite population growth, the government has taken steps to make Bangladesh food-sufficient before that estimated time.

The expansion of modern agricultural farming practices, such as use of High Yielding Variety (HYV) seed together with intensified cultivation, is needed to ensure food for all, which increases demand for fertilizers. It is therefore necessary to ensure timely supply of both organic and chemical fertilizers to meet the nutritional demand of these varieties. Use of chemical fertilizer is increasing with rising food-production demand; urea alone is the highest-used fertilizer in the country's agriculture. In FY2017-18, total fertilizer used was 50.93 lakh MT, of which urea fertilizer was 24.27 lakh MT. Development support in fertilizer was 5200.67 crore taka in FY2017-18.

The fertility status of different regions of the country varies considerably. Individual farmers have fragmented land into smaller pieces, causing wide variation in management of each piece of land; this leads to large variation in fertility level even between adjacent plots. Realizing the difficulty of generalizing fertility level, only an indicative status fertility level in different regions of the country is shown in the maps in this atlas.

There is an increasing trend of soil health degradation, attributable to higher crop removal from increasing cropping intensity, use of modern crop varieties (HYVs and hybrids), soil erosion, soil salinity, soil acidity, deforestation, nutrient leaching, and minimum manure application — mostly due to irrational human interventions. Consequently, new nutrient deficiencies arise over time: chronologically, N, P, K, S, Zn and B deficiencies have appeared. Magnesium deficiency is reported in Old Himalayan Piedmont Plain and Tista Floodplain soils. Calcium deficiency also arises in Old Himalayan Piedmont Plain and Tista, part of Brahmaputra Flood Plain, and North and Eastern Hills. There is sporadic information on Cu, Mo and Mn deficiencies in crops; deficiencies of Fe and Cl are not yet found. It is estimated that the overall N balance of Bangladesh soils is negative, the P balance is near zero, and the K balance is highly negative.

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), Background, pp.08-09]

Methodology

The present study was carried out across the cultivable land of Bangladesh (26°37'54.86" N, 87°59'59.96" E, to 20°34'1.85" N, 92°40'40.93" E), with a total study area of 14,757,000 hectares.

Soil sample points were collected from the Upazila Nirdeshika. Soil samples were subjected to physical and chemical analyses according to the SRDI module for Upazila Nirdeshika. Mapping units (MUs) were defined by grouping areas of soils at a 1:50,000 scale, according to the Upazila Nirdeshika. The upazila soil and landform map was delineated using visual interpretation of soil distribution in the area; soil-landscape relationships were established using visual interpretation of field observations, soil descriptions, and subsidiary aerial photos and topographic maps.

Due to the large amount of data available for use as covariates in modelling, a data-mining technique was used to select the most suitable dataset as an optimal set of predictors to run the model with the lowest error. Generally, smaller neighborhoods or a minimum number of points are used when the phenomenon has a great amount of variation.

Covariates were ranked by importance, and the top twelve most important were selected for soil mapping. The selection procedure combined Inverse Distance Weighting (IDW) with correlation analysis to remove covariates that correlated too highly with others. IDW interpolation determines cell values using a linearly weighted combination of a set of sample points, with weight as a function of inverse distance; the surface being interpolated should be that of a locationally dependent variable.

Workflow for Developing Digital Soil Mapping (DSM)

The source presents this as a flowchart (Fig. 1); the process steps, in sequence, are:

- 1 Compilation of Soil Legacy Data
- 2 Generation of Spatial Information (using outer boundary, latest river shape, area of Sundarban, reserved forest)
- 3 Covariate Selection
- 4 Spatial Interpolation and Regression (IDW) → produces a Raster Map
- 5 Reclassify (through value of class)
- 6 Conversion (Raster to Feature)
- 7 Overlay Execution (Erase and Update)
- 8 Generalization
- 9 Validation and Uncertainty Estimation
- 10 → Final output: Digital Soil Mapping (DSM)

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), Methodology, pp.10-12, Fig. 1]

National Soil Fertility Status Tables (2020)

The following 12 tables give the area (hectares) and percentage of arable land nationally falling into each fertility/status class, for each soil parameter mapped in this atlas. Percentages are of total mapped arable land (~8,586,860-8,586,864 ha across tables; minor ~1 ha differences between tables are as printed in the source). The corresponding choropleth maps and pie charts in the source are visual-only and are not reproduced here.

Table 1 — Soil Reaction (pH) Status

Fertility / Status Class	Area (ha)	% of Arable Land
Very Strongly to Strongly Acidic	3,921,994	45.67
Slightly Acidic	2,182,698	25.42
Neutral	1,005,558	11.71
Slightly Alkaline	1,459,510	17.00
Strongly Alkaline	17,039	0.20
Very Strongly Alkaline	65	Trace
Total	8,586,863	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.14]

Table 2 — Soil Organic Matter Status

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	2,990,355	34.83
Medium	5,082,396	59.19
High	393,512	4.58
Very High	120,601	1.40
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.16]

Table 3 — Phosphorus Status

Loamy to Clayey Soils of Upland Crops

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	4,822,353	56.16
Medium	1,781,569	20.75
Optimum	887,376	10.33
High to Very High	109,566	12.76
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.18]

Table 4 — Phosphorus Status*Loamy to Clayey Soils of Wetland Rice Crops*

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	4,316,455	50.27
Medium	1,840,200	21.43
Optimum	1,063,152	12.38
High to Very High	1,367,056	15.92
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.20]

Table 5 — Potassium Status*Loamy to Clayey Soils of Upland Crops*

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	5,139,339	59.85
Medium	2,018,590	23.51
Optimum	805,010	9.37
High to Very High	623,925	7.27
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.22]

Table 6 — Potassium Status*Loamy to Clayey Soils of Wetland Rice Crops*

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	3,712,252	43.23
Medium	2,650,264	30.87
Optimum	1,147,473	13.36
High to Very High	1,076,875	12.54
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.24]

Table 7 — Sulphur Status*Loamy to Clayey Soils of Upland Crops*

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	3,061,223	35.66
Medium	1,927,100	22.44
Optimum	1,265,155	14.73
High to Very High	2,333,386	27.17
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.26]

Table 8 — Sulphur Status*Loamy to Clayey Soils of Wetland Rice Crops*

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	3,985,000	46.41
Medium	1,799,788	20.96
Optimum	1,305,808	15.21
High to Very High	1,496,268	17.43
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.28]

Table 9 — Zinc Status

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	6,769,975	78.84
Medium	990,510	11.54
Optimum	399,444	4.65
High to Very High	426,935	4.97
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.30]

Table 10 — Boron Status

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	2,642,975	30.78
Medium	2,350,285	27.37
Optimum	1,365,198	15.90
High to Very High	2,228,434	25.95
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.32]

Table 11 — Calcium Status

renumbered from source's "Table 10" — see title-page note

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	2,106,053	24.53
Medium	1,311,470	15.27
Optimum	1,014,359	11.81
High to Very High	4,154,982	48.39
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.34]

Table 12 — Magnesium Status

renumbered from source's "Table 10" — see title-page note

Fertility / Status Class	Area (ha)	% of Arable Land
Very Low to Low	1,056,740	12.31
Medium	839,835	9.78
Optimum	895,933	10.43
High to Very High	5,794,356	67.48
Total	8,586,864	100.00

[Source: Soil Fertility Atlas Bangladesh 2020, Soil Resource Development Institute (SRDI), Ministry of Agriculture (ISBN 978-984-34-9895-3), p.36]